
CHAPTER 5

Perceived Isolation

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SOCIAL ISOLATION

Most structures and processes of the mind operate outside awareness, with only the end products reaching awareness—and then only sometimes (LeDoux, 2000; Wilson & Bar-Anan, 2008). But clearly we know a great deal about the mind from what we experience through our senses. It is just common sense that we know the shape or color of an object from simply seeing it. But this common-sense notion is incorrect. Our color perception is not simply a function of the cones in the eye responding to the wavelength of the incoming light, but rather it is the result of cortical computations that have evolved to ensure color constancy under varying conditions of luminosity (Shevell & Kingdom, 2008). Visual illusions are but one set of phenomena showing that “self-evident” truths about how the mind works can be absolutely false.

The science of the mind is not unique in this regard, however. As the historian Daniel Boorstin (1983) noted:

Nothing could be more obvious than that the earth is stable and unmoving, and that we are the center of the universe. Modern Western science takes its beginning from the denial of this commonsense axiom. . . . Common sense, the foundation of everyday life, could no longer serve for the governance of the world. When “scientific” knowledge, the sophisticated product of complicated

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instruments and subtle calculations, provided unimpeachable truths, things were no longer as they seemed. (p. 294)

Just as we once thought that we were the center of the universe, we have long regarded the mind as a solitary, independent entity. After all, the brain—the organ of the mind—is housed deep within the cranial vault, where it is protected and isolated from others. Looking out onto the social world, it is self-evident that people are distinct, independent entities with nothing binding them together. But this, too, is an illusion. We may revere the rugged individualists, but we are fundamentally social organisms. We are born to the most prolonged period of abject dependency of any mammal. For the species to survive, human infants must instantly engage their parents in protective behavior, and the parents must care enough about their offspring to nurture and protect them. Even once grown we are not particularly splendid physical specimens. Other animals can run faster, see and smell better, and fight more effectively than we can. Our major evolutionary advantage is our brain and ability to communicate, remember, plan, and work together. Our survival depends on our *collective* abilities, not our individual might.

The Social Brain

Humans were not the first bipedal creatures or the first to use tools, but humans—apparently uniquely—contemplate the history of the earth, the reach of the universe, the origin of the species, the genetic blueprint of life, and the physical basis of their own unique mental existence. None of this would be possible if not for the work of others shared among contemporaries and passed down through the ages (Cacioppo et al., 2007). The social nature of the human species is not simply an add-on, either. It has fundamentally shaped the evolution of our biological design, including the rapid increase in neocortical connectivity and intelligence.

According to the social brain hypothesis, deducing better ways to find food, avoid perils, and navigate territories has adaptive value for large mammals, but the complexities of these ecological demands pale by comparison to the complexities of social living (e.g., Dunbar & Shultz, 2007). The demands of social living include learning by social observation; recognizing the shifting status of friends and foes; anticipating and coordinating efforts between two or more individuals; using language to communicate, reason, teach, and deceive others; orchestrating relationships, ranging from pair bonds and families to friends, bands, and coalitions; navigating complex social hierarchies, social norms, and cultural developments; subjugating self-interests to the interests of the pair bond or social group in exchange for the possibility of long-term benefits; recruiting support to sanction individuals who violate group norms; and doing all this across time frames that stretch from the distant past to multiple possible futures (Dunbar, 2003; Dunbar & Shultz, 2007). Such hypotheses are difficult to test but recent evidence supports the view. In troops of baboons, Silk, Alberts, and Altmann (2003) demonstrated that a composite index of sociality was highly correlated

with infant survival. And in cross-species comparisons by Dunbar and Schultz (2007), results indicate that the evolution of large and metabolically expensive brains is more closely associated with social than ecological complexity.

Living on the Social Perimeter

Social species, by definition, create emergent organizations beyond the individual—structures ranging from dyads and families to institutions and cultures. These emergent structures evolved hand in hand with neural, hormonal, and genetic mechanisms to support them because the consequent social behaviors helped these organisms survive, reproduce, and care for offspring sufficiently long that they too reproduced. Social isolation for such species is not only dangerous, it is deadly (see Cacioppo & Hawkley, 2009, for a review).

In the case of our species, the presence of others is not as important as the meaningful connection with others. Physical isolation can contribute to feelings of isolation (i.e., loneliness), but people can also feel lonely in a marriage, family, or crowd. Loneliness is the pain that one feels when one's need for connection is not being met, and one can feel that anywhere—even when surrounded by friends or family. Indeed, it is not the number of friends or contacts that predicts loneliness, it is the quality of those relationships (Hawkley et al., 2008; Wheeler, Reis, & Nezlek, 1983; see Cacioppo & Hawkley, 2009, for a review). Included in the documentation of these associations are longitudinal as well as cross-sectional studies, and evidence that the association holds even when controlling for other risk factors such as depression.

Social interactions and relationships also have a fundamental role in both the development and treatment of mental disorders. Loneliness is associated with suicidal ideation and behavior (e.g., Rudatsikira, Muula, Siziya, & Twa-Twa, 2007; Stravynski & Boyer, 2001) and depression (e.g., Heikkinen & Kauppinen, 2004), with cross-lagged panel analyses indicating that loneliness leads to increases in depressive symptomatology (Cacioppo, Hawkley, & Thisted, 2010). A defining feature of autism spectrum disorder is impairment of social cognition and affiliation (Lim, Bielsky, & Young, 2005), and schizophrenia is characterized by a deficit in pleasurable responses to social engagement (Horan & Blanchard, 2003a, 2003b). Another syndrome, known as Williams-Beuren syndrome, is characterized by hypersociability, combined with increased nonsocial anxiety (Meyer-Lindenberg et al., 2005); and psychopathy is characterized by a lack of empathy and the formation of transient and exploitive interpersonal relationships (Harpur, Hare, & Hakstian, 1989). Many deleterious health behaviors, ranging from alcoholism to obesity, have strong social components as well (Hawkley, Thisted, & Cacioppo, 2009). Understanding the connections among minds will permit better diagnosis and treatments of the chronic health problems that our society faces.

The evidence has built to the point that loneliness can be considered a serious risk factor for poor health, joining more established factors such as obesity or smoking (e.g., Holt-Lunstad, Smith, & Layton, 2010). But for those risk factors,

there are methods to help people stop smoking or reduce their weight, thereby decreasing their risk of disease. However, neither the details of the mechanism linking loneliness to infirmity nor the best strategy for reducing a person's isolation are clear (Masi, Chen, Hawkley, & Cacioppo, 2011). Finding answers to these questions soon is critical to the health and health care costs of our nation. A recent national survey of adults 45 years of age and older found that the number of individuals scoring high in loneliness has increased by nearly 50% since 2001, and the leading wave of the "baby boom" generation reached retirement age in 2011 (Edmonson, 2010).

LONELINESS AS A SIGNAL

Although the presence of and interactions with others is associated with lower levels of loneliness (e.g., Hawkley et al., 2008), they are not sufficient to produce low levels of loneliness (see review by Masi et al., 2011). For instance, when they arrive at college, freshmen are sharing housing and are surrounded by hundreds of other students—yet on average their feelings of loneliness are heightened by the fact that they have severed their normal ties with friends and family, in many instances for the first time in their lives (Shaver, Furman, & Buhrmester, 1985). Similarly, a bereaved spouse may suffer the searing pain of loneliness despite the increased presence of more supportive family and friends.

But it does not require a life-changing event to find oneself lonely in the midst of an active social life. The frenetic pace with which many in contemporary society move from work to family obligations to social activities can leave one feeling at a loss for any meaningful human contact (Edmonson, 2010). About a quarter-century ago, when Americans in a national survey were asked about the number of confidants they had, the most frequent response was three. This question was asked again a few years ago and the most frequent response was zero (McPherson, Smith-Lovin, & Brashears, 2006). This response is consistent with the robust finding that it is the quality of our social relationships—not the quantity—that is an essential part of what it takes for us to be healthy and happy (Cacioppo et al., 2008).

No Redeeming Features?

Loneliness was initially characterized in the social sciences as an aversive state with no redeeming features (Weiss, 1973), and as a state barely different from general negativity or depressed mood (cf. Booth, 2000). Our research suggests a very different depiction of loneliness. We have adopted the perspective of loneliness as a state that has evolved as a signal to change behavior—very much like hunger, thirst, or physical pain—that helps one avoid damage and promote the transmission of genes to the gene pool. In the case of loneliness, the signal is a prompt to renew the connections that we need to survive and prosper. Specifically, early in our history as a species, we survived and prospered only by

banding together—in couples, in families, in tribes—to provide mutual protection and assistance. Loneliness evolved like any other form of pain. The social pain of loneliness prompted us to renew the connections we needed to ensure survival and promote social trust, cohesiveness, and collective action. Hunger, if ignored, ultimately reduces a person’s ability in the wild to find and capture food. Loneliness too, if ignored, can have damaging effects that make it more difficult for an individual to escape its grips. Although loneliness may *feel* like it has no redeeming features, it is an aversive state that has evolved as a signal to change behavior to promote the survival of our genes (Cacioppo, Hawkley, & Correll, in press).

Evolutionary versus Contemporary Time

In evolutionary time social groupings were relatively small and stable, and the pain of loneliness may have served both to promote the social connection necessary for the survival of the genes and as a deterrent to selfish actions that were detrimental to the group. Loneliness in contemporary society, however, may affect human cognition in sometimes maladaptive ways (see Figure 1). Specifically, feeling socially isolated can trigger implicit hypervigilance for social threats, which in turn produces attentional, confirmatory, and memorial biases. Accordingly, lonely individuals are more likely to attend to and construe their social world as threatening, hold more negative social expectations, and remember more negative social events than are non-lonely individuals.

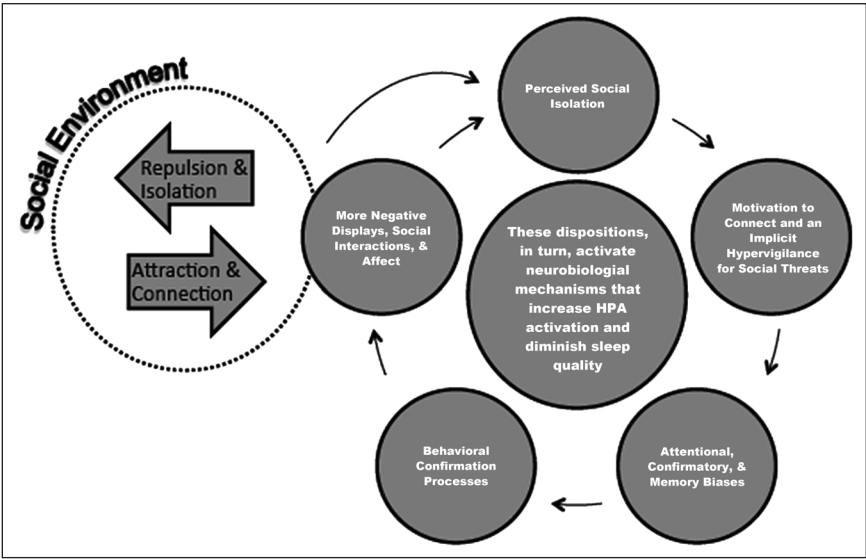


Figure 1. The effects of loneliness on human cognition. Reprinted from “Perceived Social Isolation and Cognition” by J. T. Cacioppo & L. C. Hawkley, 2009, *Trends in Cognitive Sciences*, 13, p. 451. Copyright 2009 by Elsevier Science. Reprinted with permission.

These cognitions increase the likelihood that individuals engage in behavioral confirmation processes, through which they produce more negative social interactions and elicit evidence confirming that they have little personal control or social value. These dispositions, in turn, alter the nature and likelihood of social engagement and activate neurobiological mechanisms that increase activation of the hypothalamic pituitary adrenal axis and diminish sleep quality. Repeated or chronic activation of threat surveillance in a social context—coupled with diminished anabolic processes—may contribute to heightened cognitive load, diminished executive functioning, dysregulated brain and physiological systems, and broad-based morbidity and mortality (Cacioppo & Hawkley, 2009).

Loneliness also affects behavior in a variety of ways. Because it diminishes the brain's executive functioning (Cacioppo & Patrick, 2008), it is more difficult for people who feel lonely to control impulses, such as succumbing to a guilty pleasure rather than exercising or eating a healthy diet (Hawkley et al., 2009). A person made to feel lonely not only feels unhappy but also feels unsafe, feelings that activate an anachronistic survival mechanism that heightens sensitivity to threats from all sides. Consequently, people who feel lonely may reach out to connect with others but unknowingly emphasize the negative—or *perceived* negative—features of the interaction (e.g., Cacioppo & Hawkley, 2005; Hawkley, Preacher, & Cacioppo, 2007). Thus loneliness can cause people to feel and then act distantly toward others.

However, loneliness not only signals the prospect that our social connection is at risk or absent and motivates us to repair or restore the safe, collaborative social surround we need to ensure a genetic legacy; it may also provide incentives to become more compassionate and empathic members of our social species. Consider the contemporary practice of timeout. When a child acts selfishly or socially inappropriately, a common practice is to isolate the child from the others for approximately one minute per year of age. This enforced social isolation is typically painful for the child and may be associated with displays of sadness or general negative affect, which we noted above serves as an appeal for connection and support by anyone in the vicinity. Upon reintegration to the group following the timeout, the child tends to act in a more empathic, less narcissistic fashion. That is, the existence of the aversive state of loneliness can contribute to our socialization and culture.

Although timeout is a recent innovation, it is illustrative of a more general principle in which an individual in a social setting who is made to feel isolated is compelled to change his or her behavior toward others. The “silent treatment” between close social others, workplace expulsion, teasing, being excluded or ignored in social circumstances, and countless other passive or active rejection behaviors—whether enacted for virtuous or malicious reasons—are potent elicitors of social pain (Williams, Suchy, & Rau, 2009) and can also promote other-oriented social motives. Evidence of shunning and ostracism can be found in nonhuman social species, as well as in various cultures across human history.

Ostracism in most social species is associated with an early death (Williams, 2001). In humans the chronic feeling of social isolation, even when the person remains among the protective embrace of others, is associated with significant adverse mental and physical outcomes (Hawkley & Cacioppo, 2009; Patterson & Veenstra, 2010). Because humans need to be able to work together to survive and prosper, the effects of timeout, shunning, and related methods of interpersonal rejection on people's social skills (e.g., reduced selfishness, increased perspective taking and empathy) benefit the individual and contribute to social adhesion and resilience.

CHRONIC LONELINESS

Because loss of social connection is such a threat, loneliness engenders fear as well as pain. For each of us, our need for social connection and our sensitivity to social pain is biased by our individual genetic inheritance (Boomsma, Willemssen, Dolan, Hawkley, & Cacioppo, 2005). Fear-based responses, understandably, prompt us to look out for ourselves; this can interfere with the accurate perceptions that we need to effectively connect with others and the selection of the social skills or appropriate social responses in any given occasion. Thus, when a person becomes lonely, he or she can be caught in a feedback loop in which the undercurrent of danger and threat associated in evolutionary terms with social isolation can promote a form of social cognition and interaction that becomes a self-perpetuating, self-fulfilling prophecy.

Although loneliness is about 50% heritable (Boomsma et al., 2005; McGuire & Clifford, 2000), this does not mean that loneliness is determined by genes. First, an equal amount of the variance across adults in terms of their levels of loneliness is due to situational factors. What appears to be heritable is the intensity of pain felt when one feels socially isolated (Cacioppo et al., in press). Being sensitive or insensitive are each fine, but it is crucial to create a social environment that matches one's predisposition toward feeling social pain. If one is especially sensitive, then it may benefit one's health and well-being to prioritize the development and maintenance of a few high-quality relationships.

Certain situational factors heighten the risk for increased loneliness. These include loss of a meaningful relationship, feeling disrespected by others, unresolved conflict with or lacking trust in others, family problems, feeling rejected or left out, poor marital quality, infrequent contact with friends and family, few social roles, a lack of participation in voluntary organizations, and physical health symptoms and physical limitations (cf. Hawkley et al., 2008). These situational factors can diminish the extent to which a person feels connected to others, but the resulting loneliness motivates people to reconnect with others. If these attempts are unsuccessful, the feelings of loneliness become uncoupled from the situations that aroused them, leaving one feeling chronically lonely.

THE CONSTRUCTION OF LONELINESS

Are chronic feelings of loneliness constructed by individuals based on their expectations of and responses to others? To examine this question, we manipulated levels of loneliness by hypnotizing our experimental subjects (Cacioppo et al., 2006). Using precisely worded scripts, we guided our hypnotized volunteers to reexperience moments in their lives that summoned up either profound feelings of loneliness or profound feelings of social connectedness. With some individuals we induced loneliness in their first hypnotic state and social connectedness in their second; with the others the order was reversed. Before and after each hypnotic induction, we administered the R-UCLA Loneliness Scale to ensure that the hypnosis had induced the desired emotional state.

After the hypnotic induction had produced feelings of loneliness or social connectedness, we administered the same psychological tests that we had administered previously to another group of students. As displayed in Figure 2, the results were a match. The students who were tested for naturally occurring feelings of loneliness (or the lack thereof) exhibited specific levels of eleven other characteristics that varied in relation to their degree of loneliness. Merely by manipulating feelings of loneliness and social connectedness, we had managed to produce corresponding appearances by—with corresponding levels of intensity from—all the other players in the drama. Loneliness, then, changed a variety of related dispositional orientations.

The similarity in the profiles in the top and bottom panels of Figure 2 means that lonely and non-lonely individuals are not different breeds. Anyone can succumb to loneliness and (along with it) all the other characteristics that travel as its entourage. The surveys of the students—as well as the manipulations by way of hypnosis—showed the effect of loneliness on thoughts, moods, self-regulation, and even personal characteristics such as shyness and self-esteem.

LONELINESS AND DEPRESSION

Loneliness has often been thought to be little different than depression. The experimental manipulation of loneliness showed that it increased dysphoric feelings as well (see Figure 2), but then loneliness and depressive symptomatology are both aversive, unpleasant states. However, loneliness is a debilitating

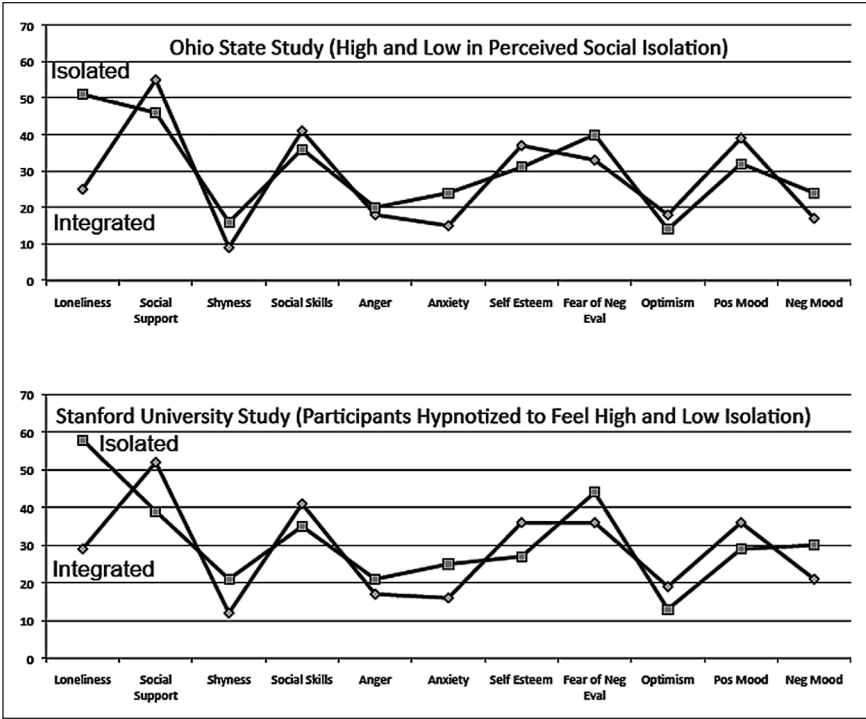


Figure 2. Top panel: the characteristics of people who are high versus low in loneliness. The solid line represents the characteristics of college students who score among the top fifth percentile in terms of loneliness (high lonely), whereas the dotted line represents the characteristics of their fellow college students who score among the bottom fifth percentile in loneliness (low lonely). These data show that individuals high—in contrast to low—in loneliness report significantly lower levels of social support, higher levels of shyness, poorer social skills, higher anger, higher anxiety, lower self-esteem, higher fear of negative evaluation, lower optimism, lower positive mood, and higher negative mood. Bottom panel: the characteristics of people when they are induced experimentally to be high versus low in loneliness. The solid line represents the characteristics of college students when they were induced to feel high in loneliness, whereas the dotted line represents the characteristics of these same students when they were induced to feel low in loneliness. These data show that when individuals feel high—in contrast to low—in loneliness, they report significantly lower levels of social support, higher levels of shyness, poorer social skills, higher anger, higher anxiety, lower self-esteem, higher fear of negative evaluation, lower optimism, lower positive mood, and higher negative mood.

psychological condition characterized by a deep sense of social isolation, emptiness, worthlessness, lack of control, and personal threat. Feeling lonely means that one wants to be close to others but either has no one in whom one can confide, depend on, or trust, or feels dissatisfied with whom one can confide in,

depend on, or trust. Feeling lonely, therefore, is not only an unhappy social environment, it is also a profoundly unsafe social environment. Feeling depressed, by contrast, means that one generally feels sad and lethargic.

Thus loneliness refers to how people feel about their social connections, whereas depression refers to how people feel generally (Weiss, 1973). Because loneliness can lead to depression, a person may feel both lonely and depressed. However, one can be depressed for other reasons than loneliness, and loneliness does not always lead to depression. Many college students experience feelings of loneliness and sadness about the loss of contact with their friends and family, but the loneliness motivates the students to make new friends and they escape the clutches of loneliness before it leads to depression.

The conflation of loneliness with depressive symptomatology in the literature is extensive, however, and epidemiological measures of depressive symptomatology (the Center for Epidemiologic Studies Depression Scale; CES-D) include an item about feeling lonely. Empirical studies that have investigated this association in normal samples have found items from the loneliness and CES-D scale to load on separate factors. For instance, we (Cacioppo et al., 2006) performed a factor analysis of the responses given by a population-based sample of 229 middle-aged and older adults to the twenty-item UCLA Loneliness Scale and the twenty-item CES-D. Nine factors were extracted, and each factor was largely scale-specific: five of the factors consisted of loneliness items, and four others consisted of CES-D items—with the exception of the “I feel lonely” item, which loaded more strongly on a loneliness factor. An additional factor analysis in which the number of factors was constrained to two also showed a clear separation of the two factors, as indicated by scale-specific items on each factor. Item loadings showed an exact correspondence with the scale from which each item was derived. That is, the items from the loneliness scale loaded on the loneliness factor, and the items from the depressed affect scale loaded on the separate depressive symptomatology factor.¹

The existing longitudinal research further supports the distinction between loneliness and depressive symptomatology, where loneliness has been found to predict increases in depressive symptomatology above and beyond what can be explained by initial levels of depressive symptomatology. For instance, a five-year structural equation analysis in a population-based sample revealed that loneliness in Year *n* predicted increases in depressive symptomatology in Year *n*+1

1. The results of factor analyses are, of course, dependent on the particular mix of items subjected to the analysis, so in a second study, the responses of 2,525 young adults to the UCLA loneliness scale and the Beck Depression Inventory were subjected to factor analyses. Items on the loneliness and depressive symptomatology scales again loaded on separate factors, and the loadings of loneliness items on the depressed affect factor were very low (i.e., < 0.10), as were the loadings of depressed affect items on the loneliness factors (Cacioppo, Ernst et al., 2006).

beyond what could be predicted by depressive symptomatology in Year *n*, and this remained the case even when controlling statistically for covariates including perceived stress, objective stressors, hostility, and social support (Cacioppo et al., 2010).

In sum, it is a sign of loneliness if people feel socially isolated from or rejected by the people around them, or if they are unhappy about the social relationships they do have. It is a sign of depression, however, if people feel fatigued, hopeless, helpless, and sad more generally, and if even pleasant nonsocial events in one's life (e.g., a beautiful sunset) are no longer a source of joy.

THE PREVALENCE OF LONELINESS

Social and demographic trends are placing an increasing number of adults at risk for loneliness, an established risk factor for physical and mental illness. In 1985, when researchers asked a cross-section of the American people, "How many confidants do you have?" the most common response to the question was three. In 2004, when researchers asked again, the most common response—made by 25% of the respondents—was none. One-quarter of these twenty-first-century Americans said that they had no one at all with whom to talk openly and intimately (McPherson et al., 2006). This finding was met with a great deal of skepticism, in part because of methodological differences between the 1985 and 2004 surveys. However, in 2010 the American Association of Retired Persons (AARP) conducted a national survey to determine the level of loneliness in adults 45 years of age and older in the United States (Edmonson, 2010). Results revealed a 50% increase over the past decade in the number of people who reported feeling lonely.

People used to think that infants needed only their physical needs to be cared for, but we then learned that human contact and affection were also important for most infants to do well (cf. Holt-Lunstad et al., 2010). We see a parallel in today's views toward older adults. Our society pays a great deal of attention to the financial and medical conditions in which older adults live, but we pay little or no attention to their social environment—whether they feel socially isolated or have people with whom they can interact, whom they trust, and about whom they care. Loneliness is still regarded by most people as a personal deficit or weakness. Because loneliness is stigmatized, those who are afflicted by loneliness tend to deny it, ignore it, or "tough it out," and those who do not feel lonely regard it as the other person's problem. The evidence has built to the point that loneliness could be considered a serious risk factor for poor health, joining more established factors such as obesity or smoking, and the AARP survey draws attention to how prevalent the problem of loneliness has become.

REDUCING LONELINESS

The correct strategy for reducing a person's loneliness is not obvious. Is it simply a matter of surrounding lonely persons with people, giving them more opportunities to socialize? Do they need help developing social skills? Or does a lonely person need a sort of cognitive tune-up, a realignment to break out of the cycle of negative social thoughts and perceptions? If we know that loneliness is involved in health problems, the next question is what we can do to mitigate it.

In the past few decades, approximately fifty intervention studies have been conducted to reduce loneliness. These intervention studies can be characterized as adopting one of four strategies: (a) improving social skills; (b) enhancing social support; (c) increasing opportunities for social contact; and (d) addressing maladaptive social cognition. We performed an integrative meta-analysis of loneliness reduction interventions to quantify the effects of each strategy and examine the potential role of moderator variables (Masi et al., *in press*). Results revealed that single group pre-post and nonrandomized comparison studies yielded larger mean effect sizes relative to randomized comparison studies. Among studies that used the latter (most rigorous) experimental design, the most successful interventions addressed maladaptive social cognition.

This result is consistent with our model of loneliness as regulatory loop, in which lonely individuals have increased sensitivity to and surveillance for social threats, preferentially attend to negative social information (Cacioppo, Norris, Decety, Monteleone, & Nusbaum, 2009), remember more of the negative aspects of social events (Duck, Pond, & Leatham, 1994), hold more negative social expectations (Cacioppo & Hawkley, 2005), and are more likely to behave in ways that confirm their negative expectations (see Figure 1). This loop has short-term self-protective features, but over the long term it heightens cognitive load, diminishes executive functioning, and adversely influences physical and mental health and well-being. Among the four intervention types, addressing maladaptive social cognition most directly addresses this regulatory loop.

CONCLUSION

Cervone and Bartoszek (this volume) note that nomothetic analyses can mask idiographic effects. The research on loneliness reinforces this notion and shows that idiographic and nomothetic analyses can provide converging approaches to illuminate universals. Loneliness produces both a motivation to affiliate with others and increased hostility toward others—a seemingly paradoxical set of effects that Andersen and Przybylinski (this volume) also find in their studies of transference and significant others. Trusting others satisfies our need for social approval and provides epistemic security. The absence of such connections is both unhappy and unsafe, and the mixed motives to affiliate with and feel threatened by others can be adaptive (Cacioppo et al., *in press*), even if they also leave people largely an unintentional architect of their own social reality.

Acknowledgments

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REFERENCES

- Boomsma, D. I., Willemsen, G., Dolan, C. V., Hawkey, L. C., & Cacioppo, J. T. (2005). Genetic and environmental contributions to loneliness in adults: the Netherlands twin register study. *Behavior Genetics*, 35, 745–752.
- Boorstin, D. J. (1983). *The discoverers: A history of man's search to know his world and himself*. New York: Random House.
- Booth, R. (2000). Loneliness as a component of psychiatric disorders. *Medscape General Medicine*, 2, 1–7.
- Cacioppo, J. T., Amaral, D. G., Blanchard, J. J., Cameron, J., Carter, C., Crews, D., et al. (2007). Social neuroscience: Progress and implications for mental health. *Perspectives on Psychological Science*, 2, 99–123.
- Cacioppo, J. T., Ernst, J. M., Burleson, M. H., McClintock, M. K., Malarkey, W. B., Hawkey, L. C., et al. (2000). Lonely traits and concomitant physiological processes: The MacArthur Social Neuroscience Studies. *International Journal of Psychophysiology*, 35, 143–154.
- Cacioppo, J. T., & Hawkey, L. C. (2005). People thinking about people: The vicious cycle of being a social outcast in one's own mind. In K. D. Williams, J. P. Forgas, & W. von Hippel (Eds.), *The social outcast: Ostracism, social exclusion, rejection, and bullying* (pp. 91–108). New York: Psychology Press.
- Cacioppo, J. T., & Hawkey, L. C. (2009). Perceived social isolation and cognition. *Trends in Cognitive Sciences*, 13, 447–454.
- Cacioppo, J. T., Hawkey, L. C., & Correll, J. (in press). Perceived social isolation within personal and evolutionary timescales. In C. N. DeWall (Ed.), *Handbook of social exclusion*. New York: Oxford University Press.
- Cacioppo, J. T., Hawkey, L. C., Ernst, J. M., Burleson, M. H., Berntson, G. G., Nouriani, B., et al. (2006). Loneliness within a nomological net: An evolutionary perspective. *Journal of Research in Personality*, 40, 1054–1085.
- Cacioppo, J. T., Hawkey, L. C., Kalil, A., Hughes, M. E., Waite, L. J., Thisted, R. A., et al. (2008). Happiness and the invisible threads of social connection: The Chicago Health, Aging, and Social Relations Study. In M. Eid & R. Larson (Eds.), *The science of well-being* (pp. 195–219). New York: Guilford Press.
- Cacioppo, J. T., Hawkey, L. C., & Thisted, R. A. (2010). Perceived social isolation makes me sad: Five-year cross-lagged analyses of loneliness and depressive symptomatology in the Chicago Health, Aging, and Social Relations Study. *Psychology and Aging*, 25, 453–463.
- Cacioppo, J. T., Norris, C. J., Decety, J., Monteleone, G., & Nusbaum, H. (2009). In the eye of the beholder: Individual differences in perceived social isolation predict regional brain activation to social stimuli. *Journal of Cognitive Neuroscience*, 21, 83–92.
- Cacioppo, J. T., & Patrick, B. (2008). *Loneliness: Human nature and the need for social connection*. New York: Norton.

- Duck, S., Pond, K., & Leatham, G. (1994). Loneliness and the evaluation of relational events. *Journal of Social and Personal Relationships*, *11*, 253–276.
- Dunbar, R. I. (2003). Evolution of the social brain. *Science*, *302*, 1160–1161.
- Dunbar, R. I., & Shultz, S. (2007). Evolution in the social brain. *Science*, *317*, 1344–1347.
- Edmondson, B. (2010). All the lonely people. *AARP: The Journal*, November/December, 83.
- Harpur, T. J., Hare, R. D., & Hakstian, R. A. (1989). Two-factor conceptualization of psychopathy: Construct validity and assessment implications. *Psychological Assessment*, *1*, 6–17.
- Hawkley, L. C., Hughes, M. E., Waite, L. J., Masi, C. M., Thisted, R. A., & Cacioppo, J. T. (2008). From social structural factors to perceptions of relationship quality and loneliness: The Chicago Health, Aging, and Social Relations Study. *Journals of Gerontology, Series B: Psychological Sciences and Social Sciences*, *63*, S375–S384.
- Hawkley, L. C., Preacher, K. J., & Cacioppo, J. T. (2007). Multilevel modeling of social interactions and mood in lonely and social connected individuals: The MacArthur Social Neuroscience Studies. In A. D. Ong & M. H. Van Dulmen (Eds.), *Oxford handbook of methods in positive psychology* (pp. 559–575). New York: Oxford University Press.
- Hawkley, L. C., Thisted, R. A., & Cacioppo, J. T. (2009). Loneliness predicts reduced physical activity: Cross-sectional and longitudinal analyses. *Health Psychology*, *28*, 354–363.
- Heikkinen, R., & Kauppinen, M. (2004). Depressive symptoms in late life: A 10-year follow-up. *Archives of Gerontology and Geriatrics*, *38*, 239–250.
- Holt-Lunstad, J., Smith, T. B., & Layton, J. B. (2010). Social relationships and mortality risk: A meta-analytic review. *PLoS Medicine*, *7*(7), e1000316.
- Horan, W. P., & Blanchard, J. J. (2003a). Emotional responses to psychosocial stress in schizophrenia: The role of individual differences in affective traits and coping. *Schizophrenia Research*, *60*, 271–283.
- Horan, W. P., & Blanchard, J. J. (2003b). Neurocognitive, social, and emotional dysfunction in deficit syndrome schizophrenia. *Schizophrenia Research*, *65*, 125–137.
- LeDoux, J. E. (2000). Emotion circuits in the brain. *Annual Review of Neuroscience*, *23*, 155–184.
- Lim, M. M., Bielsky, I. F., & Young, L. J. (2005). Neuropeptides and the social brain: Potential rodent models of autism. *International Journal of Developmental Neuroscience*, *23*, 235–243.
- Masi, C. M., Chen, H. Y., Hawkley, L. C., & Cacioppo, J. T. (2011). A meta-analysis of interventions to reduce loneliness. *Personality and Social Psychology Review*, *15*, 219–266.
- McGuire, S., & Clifford, J. (2000). Genetic and environmental contributions to loneliness in children. *Psychological Science*, *11*, 487–491.
- McPherson, M., Smith-Lovin, L., & Brashears, M.E. (2006). Social isolation in America: Changes in core discussion networks over two decades. *American Sociological Review*, *71*, 353–375.
- Meyer-Lindenberg, A., Hariri, A. R., Munoz, K. E., Mervis, C. B., Mattay, V. S., Morris, C. A., et al. (2005). Neural correlates of genetically abnormal social cognition in Williams syndrome. *Nature Neuroscience*, *8*, 991–993.
- Patterson, A. C., & Veenstra, G. (2010). Loneliness and risk of mortality: A longitudinal investigation in Alameda County, California. *Social Science and Medicine*, *71*, 181–186.

- Rudatsikira, E., Muula, A. S., Siziya, S., & Twa-Twa, J. (2007). Suicidal ideation and associated factors among school-going adolescents in rural Uganda. *BMC Psychiatry*, 7, 67.
- Shaver, P. R., Furman, W., & Buhrmester, D. (1985). Transition to college: Network changes, social skills, and loneliness. In S. Duck & D. Perlman (Eds.), *Understanding personal relationships: An interdisciplinary approach* (pp. 193–219). London: Sage.
- Shevell, S. K., & Kingdom, F. A. (2008). Color in complex scenes (pp. 143–166). *Annual Review of Psychology*, 59, 143–166.
- Silk, J. B., Alberts, S. C., & Altmann, J. (2003). Social bonds of female baboons enhance infant survival. *Science*, 302, 1231–1234.
- Stravynski, A., & Boyer, R. (2001). Loneliness in relation to suicide ideation and parasuicide: A population-wide study. *Suicide and Life-Threatening Behavior*, 31, 32–40.
- Weiss, R. (1973). *Loneliness: The experience of emotional and social isolation*. Cambridge, MA: MIT Press.
- Wheeler, L., Reis, H., & Nezlek, J. (1983). Loneliness, social interaction, and sex roles. *Journal of Personality and Social Psychology*, 45, 943–953.
- Williams, K. D. (2001). *Ostracism: The power of silence*. New York: Guilford Press.
- Williams, P. G., Suchy, Y., & Rau, H. K. (2009). Individual differences in executive functioning: Implications for stress regulation. *Annals of Behavioral Medicine*, 37, 126–140.
- Wilson, T. D., & Bar-Anan, Y. (2008). The unseen mind. *Science*, 321, 1046–1047.